

Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads

Organization: MathWorks

Department: MathWorks Engineering

1. Project Executive Summary & Core Objective

Design and simulate an end-to-end closed-loop autonomous driving system capable of real-time perception, multi-agent motion prediction, behavior decision-making, and adaptive collision-free path planning on **unstructured Indian roads** where standard lane markings are absent, traffic participants (auto-rickshaws, two-wheelers, cyclists, stray cattle, pedestrians) exhibit non-lane behavior, and road conditions feature potholes and uncertain boundaries.

CLOSED-LOOP SYSTEM PIPELINE:

Sensors (Camera + LiDAR + Radar) → Perception & Fusion → Object Tracking & Motion Prediction → Behavior Decision (Stateflow) → Adaptive Local Path Planner (Frenet / TEB) → Vehicle Controller (MPC / Pure Pursuit) → Vehicle Dynamics (Bicycle Model) → Updated Scenario Observations

2. Master Dataset Directory (Direct Links & Details)

Clickable direct links for all datasets categorized by traffic, animals, humans, and point clouds:

Dataset & Type	Classes & Key Features	Size / Format	Official Download Link
Indian Driving Dataset (IDD) Primary Indian Roads	Auto-rickshaws, motorcycles, riders, pedestrians, stray animals , trucks, unstructured road boundaries.	IDD-Lite: ~15 GB Full: ~100 GB YOLO / JSON	idd.insaan.iiit.ac.in
Roboflow Indian Cattle Animals on Road	Cows, stray cattle, buffaloes, dogs, goats on Indian asphalt & dirt roads. Pre-annotated.	~500 MB – 2 GB Direct YOLOv8 / YOLO11	universe.roboflow.com
Roboflow Road Animals Stray Animals & Dogs	Stray dogs, street animals crossing paths, sudden obstacle animal datasets.	~1 GB YOLOv8 PyTorch	universe.roboflow.com
nuScenes Dataset Multi-Sensor Perception	360° Cameras, 32-beam LiDAR, Radars. Rich 3D bounding boxes & pedestrian tracks .	Mini: ~4 GB Full: ~300 GB nuScenes format	nuscenes.org/download
KITTI 3D Vision Benchmark Camera & LiDAR	Velodyne HDL-64E LiDAR point clouds, stereo cameras, calibrated 3D labels for cars & pedestrians.	~12 GB (3D Object Detection) Bin/Txt	cvlibs.net/kitti
BDD100K Dataset Crowded Pedestrians	100,000 driving videos, dense pedestrian crowds, night & rainy weather conditions.	Images: ~20 GB Labels: ~100 MB	bdd-data.berkeley.edu
Mendeley Indian Traffic Indian Vehicles	Real-world Indian mixed traffic photos & video clips (buses, trucks, auto-rickshaws, bikes).	~2–5 GB Images/Videos	data.mendeley.com
Roboflow Pothole Detection Road Damage & Potholes	Indian road potholes, cracks, speed breakers, and unmarked road edge anomalies.	~800 MB YOLOv8 format	universe.roboflow.com

3. Pre-trained Models & Code Repositories

Module Domain	Model / Algorithm	GitHub Repo & Documentation Link	Installation & Execution Command
2D Vision Perception	YOLOv8 / YOLO11 Ultralytics State-of-the-Art	github.com/ultralytics/ultralytics	pip install ultralytics yolo detect train data=idd.yaml model=yolov8m.pt epochs=50
3D LiDAR Perception	MMDetection3D PointPillars & CenterPoint	github.com/open-mmlab/mmdetection3d	pip install openmim mim install mmdet3d
Multi-Object Tracking	ByteTrack & BoT-SORT Real-time association	github.com/ifzhang/ByteTrack	pip install lap yolo track model=best.pt tracker='bytetrack.yaml'
Motion Prediction	Trajectron++ Multi-agent trajectory forecast	github.com/StanfordASL/Trajectron-plus	python train.py --eval_every 10 --dataset nusenes
Path Planning & Control	PythonRobotics Frenet, Hybrid A*, MPC	github.com/AtsushiSakai/PythonRobotics	python frenet_optimal_trajectory.py
MATLAB Simulation	Automated Driving Toolbox Stateflow & DrivingScenario	mathworks.com/automated-driving	drivingScenarioDesigner roadrunner

4. Mandatory Test Scenarios & Simulation Specs

Scenario 1: Unmarked Village Road: Narrow village road with missing boundary lines, oncoming tractor/bus, parked two-wheelers, and occasional pedestrian walking on asphalt edge.

Scenario 2: Uncontrolled Urban Intersection: Busy 4-way intersection without traffic lights or standard right-of-way. Vehicles, bikes, and auto-rickshaws cutting across irregularly.

Scenario 3: Highway Merge with Slow Traffic: Autonomous vehicle merging or overtaking slow-moving overloaded trucks and auto-rickshaws under uncertain headway gaps.

Scenario 4: Dense Market Area with Mixed Traffic: Pedestrians jaywalking, roadside pushcart vendors, parked bicycles, shop-front encroachments, and high lateral movement.

Scenario 5: Sudden Cattle-Crossing Event: Cow / stray dog stepping onto the path with low Time-to-Collision (TTC), requiring immediate risk evaluation, local evasion or emergency braking.

5. Powerful PC Hardware & Environment Setup Guide

RECOMMENDED HARDWARE SPECIFICATIONS:

- **GPU:** NVIDIA RTX 3060 / 3080 / 4070 / 4080 / 4090 (Minimum 8GB+ VRAM, recommended 12GB+ for 3D LiDAR & YOLO)
- **CPU:** Intel Core i7 / i9 (12th+ Gen) or AMD Ryzen 7 / 9 (8+ Cores for parallel simulation)
- **RAM:** 32 GB DDR4/DDR5 (64 GB ideal for point cloud processing & RoadRunner 3D rendering)
- **Storage:** 1 TB NVMe SSD (Datasets: IDD ~15-100GB, nuScenes ~50-300GB, BDD100K ~20GB)

QUICK CONDA & PYTORCH CUDA SETUP SCRIPT:

```
# 1. Create Virtual Environment
conda create -n sih2026 python=3.10 -y && conda activate sih2026
# 2. Install PyTorch with CUDA 12.1
pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu121
# 3. Install Perception, Vision & Tracking Packages
pip install ultralytics opencv-python numpy scipy matplotlib open3d filterpy lap reportlab
```

6. Evaluation Metrics & Deliverables Checklist

Evaluation Metrics: (1) Scenario Completion Rate (%), (2) Collision & Near-Collision Counts, (3) Minimum Obstacle Clearance (m), (4) Replanning Latency (ms), (5) Path Smoothness & Jerk (m/s^3), (6) Average & Max Lateral Error (m), (7) Emergency Braking Frequency.

Final Deliverables Checklist:

- ☐ Working closed-loop MATLAB & Simulink simulation pipeline.
- ☐ 2 Detailed 3D RoadRunner scenes (Village Road + Urban Unmarked Junction).
- ☐ 5 Validated Indian road scenarios with trajectory plots.
- ☐ Perception (YOLO + PointPillars) and Tracking models with weights.
- ☐ Technical Report, Demonstration Video, and Reproducible Source Code.